1.1.1 Molecular Potentials

In practice, when designing a classical potential energy function in the empirical approach, the modeller simply works within a defined framework using parameters and prescribed functional forms to match the properties of the system to experimental data (or data obtained from quantum mechanical modelling of small systems). In this subsection, we briefly survey a few of the potential energy functions used in common molecular modelling applications.

In the most common situations, the potential energy function consists of a sum of 2-body, 3-body and/or 4-body terms,

$$U_{ij}(\boldsymbol{q}_i, \boldsymbol{q}_j), \qquad U_{ijk}(\boldsymbol{q}_i, \boldsymbol{q}_j, \boldsymbol{q}_k), \qquad U_{ijkl}(\boldsymbol{q}_i, \boldsymbol{q}_j, \boldsymbol{q}_k, \boldsymbol{q}_l),$$

where $\boldsymbol{q}_i$ is the position vector of atom i such that $(q_{i,x}, q_{i,y}, q_{i,z}) = \boldsymbol{q}_i \in \mathbb{R}^3$. The 2-body terms describe repulsion due to the impenetrability of the outer valence shell of atoms (Pauli repulsion), bonds between atoms which "share" electrons, and the Coulombic attractive and repulsive forces due to net charges accumulating on the atoms of bound groups. Higher body terms model the asymmetry of the wave function for atoms in bound groups.

One of the simplest systems to study is H_2^+, i.e. a system of two nuclei sharing a single electron. We suppose the two nuclei to be initially well separated with the electron density localized to one of the two and radially symmetric about it. As the internuclear separation is decreased, the charge cloud becomes elongated in the direction of the distant nucleus due to Coulombic attraction. The two nuclei will feel an effective attractive force. As the separation is reduced, the charge cloud will be shared equally by both nuclei. As the nuclei are brought still closer together, the repulsion of the nuclei due to their positive charges comes into play. The potential energy therefore initially decreases as the atoms are brought together from distance before increasing at short range. The situation is diagrammed in Fig. 1.4. The inherent quantum mechanical nature of the interaction is shown by indicating the first few discrete energy levels accessible to the H_2^+ system (source: [100]).

Similar potential energy curves can be found for other bonds. A simple potential energy function whose graph can be used to approximate the potential energy of bond dissociation is the *Morse potential*

$$\varphi_{\text{Morse}}(r) = D\left(1 - e^{-a(r-r_e)}\right)^2.$$

(See Fig. 1.5.) r_e is the location of the minimum, D gives the well depth, and a is a shape parameter that can be used to control the curvature at the minimum. In any given situation, the constants D, a, r_e can all be used to adjust the potential to approximate an effective energy graph from experiment or quantum mechanics. For example, Fig. 1.5 shows the approximation of the H_2 (not H_2^+) bond potential by a Morse curve (which has been shifted down to match the minimum energy).